



Homogeneous Propagation Framework: A Comparative Overview

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Abstract

This paper is a clarification of what this framework presently is, and what it is not. The framework recovers a substantial portion of familiar relativistic structure, but it should not be identified either with orthodox special relativity, with orthodox general relativity, or with a finished replacement for them. In the homogeneous regime, contraction, timing structure, and the one-dimensional Lorentz map are recovered in form from a propagation-first construction rather than postulated as primitive spacetime symmetry. In the gravitational regime, weak-field timing, delay, lensing, perihelion precession, and related tests are developed from displacement of the transport-supporting structure rather than from curvature of spacetime itself. The framework is therefore not SR or GR restated in ontology or derivational order, but it still overlaps the tested kinematic and weak-field observational record within its present scope. Those overlaps are real and important, and within those recovered kinematic and weak-field domains the framework is already sufficiently closed to support its present derivations. The remaining open work enters mainly when the program is pushed further, particularly toward deeper matter structure, intra-nuclear displacement sourcing from excitation-level theory, and any broader unification with nuclear-scale physics. The focus here is the framework's present status—what has been recovered, what is physically distinct, and where the remaining frontier lies. The central claim is that the framework should presently be read as an alternative derivational and ontological program with significant overlap with SR/GR in its recovered weak-field and kinematic domains, while remaining open in the deeper unificatory sense rather than in those recovered sectors themselves.

Keywords: homogeneous propagation, emergent Lorentz structure, displacement framework, weak-field gravity, ontology, theory status, open dynamics

1. Introduction

A recurring problem for alternative frameworks is that they are often sorted too quickly into the wrong category. If they recover familiar formulas, they are treated as though they must only be a relabeling of established theory. If they depart in ontology, they are often read as claiming more than has actually been shown. This framework has encountered both reactions.

The aim is not to enlarge those claims, but to place them more exactly.

On one side, the kinematic papers recover the standard contraction factor, the timing factor, and the one-dimensional Lorentz transformation in form. The weak-field displacement papers recover the Newtonian limit, the standard weak-field perihelion structure, the logarithmic Shapiro delay, light deflection with the usual coefficient, and a weak-field route toward frame-drag behavior. It is therefore reasonable to ask whether the framework is simply special and general relativity written in different language. That question cannot be answered at the level of category labels alone, because the framework's claim is not merely that it is different from SR/GR, but that it still overlaps the tested kinematic and weak-field observational record while assigning those results a different physical basis.

On the other side, the framework also makes claims that are not standard. It treats the vacuum as a physically structured medium with storage properties, treats Lorentz structure as emergent rather than primitive, and treats gravity as displacement of the transport-supporting structure rather than curvature of spacetime itself. Those are not cosmetic differences. They are differences in what the theory says the world is.

At the same time, the framework is not complete. The displacement-sourcing relation is still foundational at the present stage: the papers currently treat nuclear-scale excitation dynamics as the source-side bridge rather than deriving that bridge from a finished substrate microphysics. The full microphysical closure is not finished, and the strong-field and fully coupled dynamical sectors remain open.

The clearest reading lies between two familiar simplifications. The framework is not merely SR/GR in different words. But neither is it yet a final replacement theory with every further sector closed and derived.

A more accurate reading is simpler. The framework is best read, at its present stage, as an alternative program with three salient features:

1. broad recovery of standard relativistic structure within the framework's present scope,
2. a distinct underlying physical interpretation of that structure, and
3. explicit remaining gaps that prevent the framework from being treated as complete in a broader unificatory sense.

The aim is to make that status precise. It does not attempt to re-derive every technical result already developed elsewhere. Instead it organizes the existing work around three direct questions: what has already been recovered, in what sense that recovered structure is not just standard relativity again, and what remains genuinely open.

2. The Framework's Present Status in One Sentence

The most concise statement of the framework's present status is the following:

This framework presently recovers large parts of the special-relativistic kinematic structure and the general-relativistic weak-field observational structure, but it does so from a different ontology and derivational order, and it does not yet constitute a final closed theory. Its overlap with the tested kinematic and weak-field record is therefore real without making it identical to SR or GR.

That sentence states the central position.

Three points matter. First, the familiar structures are recovered rather than simply assumed from the start. Second, the physical interpretation assigned to those structures is not the orthodox one. Third, the framework's remaining openness lies mainly in further sectors, especially those tied to matter structure and intra-nuclear displacement sourcing, rather than in the recovered kinematic and weak-field results themselves.

Those three points organize the discussion that follows.

3. What Has Been Recovered

3.1 Kinematic structure in the homogeneous regime

The homogeneous propagation work begins from a propagation-first condition rather than from

an a priori postulate that all inertial frames are fundamentally equivalent. Within that construction, the standard contraction factor and timing factor are recovered from the directional structure of one-way and two-way propagation cost. The one-dimensional Lorentz transformation is then recovered in form from the same propagation geometry. That program should not be collapsed too quickly into a simple "Lorentzian aether" reading, because the framework's treatment of propagation includes directional structure at both transmission and reception rather than only at one end of the exchange; Appendix A explains why that distinction matters for Michelson-Morley-type objections.

This is already a significant result. It means that a structure normally treated as primitive can instead be recovered as the endpoint of a more basic propagation analysis. In the current framework, contraction is not introduced as an independent axiom, nor is time dilation treated as a separate starting law. Both arise from the same interaction-cost structure of motion through the transport-supporting medium.

The framework also extends this recovery program into signal transfer. Doppler structure, aberration, and noncollinear composition are treated as related consequences of the same propagation-based setting rather than as disconnected formula sets imported piecemeal.

3.2 Weak-field gravitational structure

The displacement program then extends the framework beyond the homogeneous regime. Here the working postulate should be read in its narrower source-side form: intra-nuclear action -- the internal dynamics of nuclear-scale excitation structure -- sources displacement of the transport-supporting structure, while the scalar mass-energy assigned to matter is the downstream bulk measure of that sourcing capacity. Changes in this displacement propagate causally at the medium speed. In the static spherical case, this yields the equilibrium displacement geometry, the cumulative displacement field, and the Newtonian limit.

From that weak-field structure, the framework develops or recovers the following familiar results:

- gravitational clock-rate structure,
- gravitational redshift,
- Shapiro delay,
- Mercury perihelion advance,
- light deflection with the usual weak-field coefficient,

- weak-field observational consistency checks such as lunar laser ranging, and
- a rotating-source extension intended to account for geodetic and frame-drag behavior.

These are not minor overlaps. They cover much of the standard weak-field phenomenology that any serious gravitational framework must meet.

3.3 Why this overlap matters

This overlap should not be minimized. A framework that cannot recover the tested relativistic structures is not competitive. The framework does recover many of them, and that is one reason to take it seriously as a research program.

But overlap in recovered formulas does not by itself settle the question of identity. Two theories may agree over a shared domain while differing in ontology, explanatory order, or extension outside that domain. The next sections therefore ask in what sense the present framework is not simply SR or GR under another name.

4. Why This Is Not Simply Special Relativity

4.1 SR takes Lorentz structure as primitive symmetry

In orthodox special relativity, Lorentz invariance is a foundational symmetry statement about spacetime and the laws of physics. The relativity principle and invariant light speed are primitive ingredients. The Lorentz transformation then expresses the symmetry structure of spacetime itself.

This framework does not start there.

4.2 The program treats Lorentz structure as recovered

In the homogeneous propagation papers, the familiar Lorentz kinematic structure appears as a recovered endpoint of a propagation-first analysis. The framework asks what contraction, timing, and mapping structure follow once propagation through the transport-supporting layer is taken as primary. Lorentz form is then obtained from that more basic construction.

That change in derivational order is significant. In one theory, Lorentz structure is a starting symmetry. In the other, it is an emergent operational structure of systems made from and

propagating through a single-speed medium.

4.3 The physical meaning of rods and clocks is different

This difference is not merely mathematical. In the framework's own interpretation, rods and clocks are not primitive ideal objects laid down in an already-given Minkowski spacetime. They are material systems whose internal closure depends on repeated interactions carried by the medium. Contraction and time dilation therefore enter as the self-consistent response of those systems to motion through the transport structure.

Put plainly, the order of explanation is reversed. The framework does not say that matter behaves Lorentz-covariantly because spacetime is Minkowskian in the first instance. It says that matter built from a single-speed propagation medium will realize a Lorentz-type operational structure.

4.4 Preferred structure versus undetectable symmetry

The framework also permits a preferred transport-supporting structure at the ontological level, while maintaining that medium-constituted observers recover the familiar relativistic observational structure. This is sharply different from the standard interpretation of SR, where no such preferred underlying structure is admitted.

That does not automatically yield a conflict at the observational level. One can have an observationally emergent Lorentz structure with a deeper preferred substrate if material measuring systems are themselves governed by the same medium. But the explanatory picture is no longer orthodox SR.

4.5 The present conclusion about SR

The point is therefore not that the framework rejects special-relativistic kinematics. Nor is it that it simply restates SR. The more accurate statement is this:

The framework reproduces major special-relativistic kinematic relations while assigning them a different derivational order and a different ontological interpretation.

The framework therefore is not simply SR restated, but a distinct account that overlaps the tested kinematic record from a different physical basis.

5. Why This Is Not Simply General Relativity

5.1 GR identifies gravity with spacetime curvature

In orthodox general relativity, gravity is not a force in the Newtonian sense but the curvature of spacetime itself. The metric is dynamical, the Einstein field equations relate geometry to stress-energy, and free motion follows geodesics of that spacetime geometry.

This framework does not identify gravity that way.

5.2 The program identifies gravity with displacement of transport structure

In the displacement papers, gravity is treated as displacement of the transport-supporting spatial structure. The framework's older shorthand was not 'matter tells spacetime how to curve,' but rather 'mass tells space how to displace, and displaced space tells mass how to move.' The sharper source-side version is narrower: nuclear-scale excitation dynamics tell transport-supporting space how to displace, and displaced transport tells matter how to move. That is a different ontology, even where the resulting weak-field coordinate-level structure overlaps standard GR.

The framework does develop an effective coordinate geometry. But this effective geometry is downstream of the displacement picture; it is not the primitive ontological object. On the present reading, geometry here is descriptive of the displaced state, not the fundamental thing being displaced. The time and radial components are modified by the displacement field, while the angular sector remains Euclidean in the present weak-field construction. That is not merely a restatement of GR's geometrical foundation.

5.3 Agreement in the weak field does not imply identity

It should also be stated carefully that weak-field agreement with standard tests does not make the theory identical to GR. The same observational structure may arise from more than one underlying picture over a restricted domain. The framework's Mercury, Shapiro, and light-deflection results are important consistency checks, but they do not erase the distinction between curvature ontology and displacement ontology.

5.4 The field-equation status is different

A second difference is that the framework has not yet reached the same formal status as GR in its field-equation sector. GR has a closed tensorial field law. This framework instead has an intra-nuclear displacement-sourcing postulate, a retarded displacement map, a wave-equation equivalent for the cumulative displacement, and a provisional rotating-source vector extension. That is a serious dynamical program, but it is not yet a finished analog of Einstein's field equations derived from a deeper action principle or substrate-complete microphysics.

5.5 The present conclusion about GR

The point can therefore be stated as follows:

The framework reproduces substantial weak-field gravitational structure associated with GR, but it does not identify gravity with curved spacetime in the orthodox sense and it does not yet possess a fully closed gravitational field theory of comparable formal completeness.

The framework therefore is not simply GR restated, but a distinct account that overlaps the tested weak-field observational record from a different physical basis.

6. The Importance of Distinguishing Recovery from Reduction

A recurring ambiguity in theory comparison concerns the difference between **recovering** a structure and **reducing** one theory to another.

To recover a structure is to derive a formula, relation, or observable pattern that matches a known theory in a certain regime. To reduce one theory to another would be to show that the first is simply the second, perhaps in altered notation, with no additional ontological or explanatory content.

The framework currently supports the first claim in many places and does not support the second.

It recovers Lorentz-form kinematics. It recovers important weak-field gravitational observables. But the framework does not reduce itself to standard SR/GR because:

1. it starts from different postulates and a different explanatory hierarchy,
2. it gives a different physical interpretation of rods, clocks, and gravitational structure,
3. it allows a substrate / transport ontology not admitted by orthodox relativity, and
4. it remains incomplete in places where SR/GR are already formally closed.

This distinction matters both rhetorically and technically. Overstating the case as 'this is just relativity again' erases the framework's actual content. Overstating it as 'this replaces relativity in finished form' erases its actual incompleteness. The correct middle statement is stronger than either caricature.

7. What Remains Open

The framework is not unfinished in any trivial sense. Naming the remaining open work should not obscure the fact that the recovered kinematic and weak-field sectors already overlap the tested record within the framework's present scope. The open items are not merely cosmetic or pedagogical. They concern foundational closure.

7.1 The displacement sourcing relation remains foundational

At present, the framework should not be summarized merely as "mass-energy sources displacement." That phrase is useful as a bulk shorthand, but it makes mass sound like an unexplained primitive. The more precise source-side statement is that intra-nuclear action -- the internal dynamics of nuclear-scale excitation structure -- sources displacement of the transport-supporting structure. The scalar mass-energy assigned to matter is then the downstream bulk bookkeeping of that sourcing capacity.

This sourcing relation remains foundational in the sense that it has not yet been reduced to a complete excitation-level or nuclear-scale account internal to the broader program. Within the recovered kinematic and weak-field gravitational domain, the framework is already operationally closed enough to produce the main derived results. The open question is not whether that sector functions, but whether intra-nuclear displacement sourcing can be derived from a deeper matter theory rather than retained as a primitive bridge.

This is the most important incompleteness to state plainly.

7.2 The full microphysical closure is not complete

The broader program aims to ground matter, propagation, and gravity in a common medium picture. But the derivation from substrate to stable excitation to intra-nuclear action to bulk mass-loading to displacement is not yet fully closed. The excitation-layer work points in that direction, but the chain is not complete enough to claim that the displacement source sector has been wholly derived from deeper first principles internal to the framework.

That distinction matters. This paper should not imply that the recovered SR/GR-level structure is somehow still waiting to be closed on its own terms. Rather, the open work begins when the framework attempts to unify that recovered structure with a deeper account of matter, excitation, intra-nuclear action, and nuclear-scale behavior.

7.3 Strong-field and fully coupled regimes remain frontier problems

Most of the present success lies in homogeneous and weak-field sectors. Strong-field behavior, genuinely nonlinear multi-body coupling, compact-object dynamics, and full self-consistent source-field backreaction remain frontier tasks rather than settled accomplishments.

7.4 Rotating-source and vector-sector development remains provisional

The rotating-source branch is promising, and its weak-field signals are suggestive, but it is not yet at the same level of closure as the scalar weak-field sector. The framework should therefore avoid treating this branch as fully settled until the derivation and normalization chain are made comparably explicit.

7.5 Why naming the open problems helps rather than hurts

A theory gains credibility by marking its unresolved sectors accurately. Here, naming the open problems does not weaken the recovered results. It sharpens them. It makes clear which claims are already supported, which remain programmatic, and where the next technical work belongs.

8. What the Framework Should Claim at the Present Stage

The framework should presently make five claims, and no stronger claim than these is needed.

8.1 First claim: recovered kinematic structure

It can claim that the homogeneous propagation program recovers the familiar contraction, timing, and Lorentz-form kinematic relations from a propagation-first starting point.

8.2 Second claim: recovered weak-field gravitational structure

It can claim that the displacement program recovers the standard weak-field observational structure across several major tests, including timing, delay, orbital precession, and lensing. That overlap should be stated as overlap with the tested record, not as identity of ontology or causal assignment with GR.

8.3 Third claim: distinct ontology

It can claim that these results arise within a framework whose ontology differs from orthodox relativity: a structured vacuum / transport medium, emergent rather than primitive Lorentz structure, and gravity as displacement rather than as fundamental spacetime curvature. The point is not merely that the framework is not SR/GR, but that this ontological difference still leaves real overlap with the observed kinematic and weak-field benchmarks in the domains already treated.

8.4 Fourth claim: incompleteness is acknowledged

It can claim that the framework is not yet a closed final theory in the broader unificatory sense. The recovered SR/GR-level kinematic and weak-field sectors are already sufficiently closed to support the present derivations. What remains postulated, provisional, or under development are the deeper intra-nuclear sourcing and excitation-level sectors needed if the framework is to absorb nuclear theory and related structure into the same program.

8.5 Fifth claim: the value of the program lies in explanatory reorganization

It can claim that the framework's value is not only in reproducing familiar formulas, but in reorganizing them into a different explanatory hierarchy that may connect kinematics, matter structure, and gravity more tightly than the standard compartmentalized picture.

These claims are already strong and do not require rhetorical inflation.

9. Misreadings to Avoid

A paper of this kind should also pre-empt several foreseeable misreadings.

9.1 Not 'SR/GR are wrong, therefore this wins'

That is not the present paper's claim. The framework is strongest where it reproduces known successful structure, not where it merely negates it.

9.2 Not 'this is only semantics'

The framework does not differ from SR/GR only in terminology. It changes the explanatory direction and physical ontology in substantive ways.

9.3 Not 'recovered weak-field overlap means full equivalence'

Agreement in the weak-field regime does not establish identity with GR, particularly when the ontology and unfinished sectors differ.

9.4 Not 'open problems mean the recovered results do not count'

Incomplete closure does not nullify already-derived overlap. It only limits how far the present claims may be extended.

10. A Useful Comparison Table

For clarity, the framework may be summarized in the following comparative form.

Topic	Orthodox SR/ GR reading	Framework reading
Light speed / propagation	Primitive relativistic postulate or spacetime property	Consequence of the medium's propagation-supporting structure
Lorentz structure	Fundamental spacetime symmetry	Emergent operational structure recovered from propagation
Rods and clocks	Primitive measuring systems in relativistic spacetime	Material closures sustained by the same medium structure
Gravity	Curvature of spacetime	Displacement of the transport-supporting structure
Weak-field observables	Consequences of metric curvature	Consequences of cumulative displacement and effective geometry
Ontology	No preferred medium required	Preferred transport-supporting structure permitted
Formal closure	Established closed theories within their own	Closed across the recovered SR/GR kinematic and weak-field domains; open mainly where the framework is extended into deeper matter structure, intra-nuclear displacement sourcing, and nuclear / excitation-level

Topic	Orthodox SR/ GR reading	Framework reading
	stated domains	theory

The table is not meant as a proof. It is a compact statement of the present difference in explanatory order and status.

11. Why This Clarification Matters for the Paper Series

The paper series benefits from this clarification in at least three ways.

First, it tells readers what sort of work they are reading. They are not being asked to choose immediately between total equivalence and total replacement. They are being asked to evaluate a structured alternative program that overlaps strongly with standard relativity where it should.

Second, it protects the stronger technical papers from being judged by a distorted standard. A paper that recovers kinematic form from a new starting point should not be criticized for not also being a full gravitational microphysical theory. A weak-field gravity paper should not be criticized for not yet closing the excitation ontology. Distinct tasks belong to distinct papers.

Third, it gives the program a cleaner research trajectory. The next work is not 'prove everything all at once.' It is to close the specific remaining gaps: derive intra-nuclear displacement sourcing more deeply, strengthen the rotating-source sector, and extend the coupled dynamics beyond the current weak-field domain.

12. Conclusion

The homogeneous propagation framework should presently be understood neither as a mere relabeling of special and general relativity nor as a finished replacement for them. It has already done more than propose a speculative vocabulary. It has recovered significant kinematic and weak-field gravitational structure from a different starting point. But it has also not yet earned the right to be described as fully closed.

That intermediate status is not something to conceal. It is the correct description of the framework's present maturity, and it is a stronger position than either of the two easy misreadings. The framework is not empty because it overlaps relativity. It is not complete because it has strong results. Both statements can be true at once.

The strongest present statement is therefore also the most disciplined one: the framework recovers substantial relativistic structure in form and observation, interprets that structure through a distinct medium-and-displacement ontology, and remains unfinished at the deeper dynamical and substrate-closure level. Read that way, it is neither 'just SR/GR again' nor 'already complete beyond SR/GR,' but an alternative program whose overlap, differences, and open problems can all be stated plainly.

Acknowledgment of Status

The purpose here is clarification of the framework's present state. It does not supersede the technical derivations in the companion papers, but states how they should presently be read together.

Appendix A. Michelson-Morley, Transmission Aberration, and Why the Framework Is Not Simple Lorentzian Aether

A likely objection to any framework with a preferred underlying transport structure is that it sounds like a return to Lorentzian aether. If that were all the present framework amounted to, then the obvious historical objection would follow immediately: simple emission through a preferred medium should generically expose absolute motion through Michelson-Morley-type optical tests. That objection matters and should be addressed directly.

The distinction advanced here is that the present framework does not treat optical propagation as a one-sided effect in which only the receiving frame contributes the relevant aberrational response. Instead, the broader propagation program treats directional effects as entering through both transmission and reception. In the present language, the emitted propagation direction is not independent of the state of the transmitting system, and the received direction is not independent of the state of the receiving system. That two-sided structure is what separates the framework from a naive Lorentzian-aether reading.

The Michelson-Morley issue can be stated at a high level as follows. In a simple preferred-medium picture, one expects an arm aligned with the absolute-motion direction to incur a different optical cost from an arm transverse to it, producing a second-order fringe shift under rotation. Standard Lorentzian rescue arguments then rely on compensating contraction and timing effects to suppress the observable difference. The present framework proposes a different route. Because propagation direction is treated as conditioned by the transmitting structure as well as by the receiving structure, the optical path comparison is not the same as in a one-sided ballistic or one-sided medium model. The relevant directional normalization already enters at launch, not only at detection.

This matters even in the near-frame or small-separation case. If absolute motion is present, then a purely reception-only aberration picture still leaves an asymmetry in how the launched beam samples the longitudinal and transverse arms. By contrast, if transmission already carries the corresponding directional adjustment, then the outbound and return geometry are altered before the final reception event is considered. In that case the Michelson-Morley null result is no longer being explained by the same mechanism usually associated with simple Lorentzian aether.

The purpose of the present appendix is not to claim that the full Michelson-Morley derivation has been completed here in historical detail. Rather, it is to mark the exact conceptual difference that Section 3.1 depends on. The present framework is not merely saying that matter contracts in a preferred frame and therefore hides the aether. It is saying that the propagation map itself is different because both emission and reception are medium-conditioned processes. That is a materially different claim.

Accordingly, the correct comparison is not between the present framework and a textbook caricature of aether theory, but between two different propagation models: one in which the optical effect is effectively one-sided, and one in which directional structure is built into both ends of the exchange. If the latter is the correct description, then Michelson-Morley no longer functions as a simple argument that any preferred underlying structure must reduce to the historically rejected form.

For present purposes, that point is sufficient. The role of the appendix is to keep the reader from collapsing the framework into the phrase "Lorentzian aether" without noticing that the optical mechanism being proposed is different at the level of propagation itself.

Appendix B. Circular-Orbit Integration as a Case Study

One useful way to show that the framework is not merely a verbal restatement of standard relativistic mathematics is to examine a case in which the choice of mathematical object matters. Continuous circular deflection provides such a case.

The issue is not whether the Thomas-precession baseline for uniform circular motion is known. It is. The issue is what quantity should be integrated when a frame is continuously steered by an external field around a circular orbit. In the accompanying note, two candidates are compared step by step around a full orbit.

The first is the external-field steering integral, denoted here by W_{ext} . The second is the finite special-relativistic Wigner-angle expression treated as though it were the infinitesimal object to be summed continuously around the orbit. The note shows that these are not interchangeable.

For a circular orbit divided into N equal steps, let the speed be constant with

$$\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}.$$

Let the direction increment per step be

$$\Delta\phi = \frac{2\pi}{N}.$$

The full-orbit Thomas-precession baseline is then

$$\Omega_T = 2\pi(\gamma - 1).$$

The external-field steering construction uses the step contribution

$$w_{\text{ext}}(\Delta\phi) = 2 \arctan \left[(\gamma - 1) \tan \left(\frac{\Delta\phi}{2} \right) \right],$$

and sums this over the orbit:

$$W_{\text{ext}}(N) = N w_{\text{ext}} \left(\frac{2\pi}{N} \right).$$

In the small-step limit,

$$\tan\left(\frac{\Delta\phi}{2}\right) \approx \frac{\Delta\phi}{2},$$

so

$$w_{\text{ext}}(\Delta\phi) \approx (\gamma - 1)\Delta\phi,$$

and therefore

$$\lim_{N \rightarrow \infty} W_{\text{ext}}(N) = \int_0^{2\pi} (\gamma - 1) d\phi = 2\pi(\gamma - 1).$$

So the external-field steering integral converges directly to the standard Thomas-precession value for continuous circular motion.

The same note then tests what happens if one instead takes the familiar finite Wigner-angle expression for two successive noncollinear boosts and treats that finite angle as the quantity to be summed step by step around the orbit. Written as a function of the step angle $\Delta\phi$, that finite expression is

$$w_W(\Delta\phi) = 2 \arctan\left(\frac{\sin(\Delta\phi)}{K + \cos(\Delta\phi)}\right),$$

with

$$K = \frac{\gamma^{-1} + \gamma}{\gamma^{-1} - \gamma}.$$

The corresponding orbit sum is

$$W_W(N) = N w_W\left(\frac{2\pi}{N}\right).$$

When expanded for small $\Delta\phi$, this finite Wigner-angle form scales linearly with a coefficient that does not reproduce $\gamma - 1$ in the continuous-steering limit used above. As a result, its full-orbit sum converges to the wrong value for the continuously forced circular case.

The uploaded note makes this numerical mismatch explicit at muon $g - 2$ speed. For

$$\beta = 0.9994, \quad \gamma \approx 28.87,$$

so the Thomas baseline is

$$\Omega_T = 2\pi(\gamma - 1) \approx 175.12 \text{ radians per orbit.}$$

The stepwise external-field integral converges to that same value. By contrast, the finite-Wigner sum converges to only about

$$W_W \approx 6.07 \text{ radians per orbit,}$$

which is roughly 3.5% of the Thomas baseline.

This example should be read narrowly. Its claim is not that special relativity fails to describe circular motion. The standard Thomas-precession result is itself part of the relativistic baseline. The narrower point is that the finite Wigner angle for discrete noncollinear boost composition is not the same mathematical object as the infinitesimal steering integrand appropriate to continuous external-field circular motion. Treating the first as though it were the second produces a clear mismatch.

That distinction matters here because it gives a concrete example of divergence in mathematical route rather than only in verbal interpretation. The framework does not merely rename standard expressions. In this case it selects a different integrable object, and that choice is what recovers the correct continuous-orbit limit directly.

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